Yang–Mills Mass-Gap Programme

Scoped Theorems, Obstructions, and the Remaining Wilson–OS Bridge

Consolidated results and proofs through the Cycle-500 freeze

Binding scientific status. This report does not contain a proof of four-dimensional Yang–Mills existence or of a positive mass gap. It does not construct the continuum theory and it is not an authorized publication. The frozen ledgers are

$$0/9, \qquad 0/18, \qquad \text{BR–!1} = 0, \qquad \text{BR–!2 unopened}, \qquad \Delta_{\mathrm{YM}} = 0.$$

Cycle 500 closed as a typed-null programme reconciliation. The mathematical results below are exact abstract theorems, fixed-regulator facts, conditional bridges, calibrations, and scoped obstruction theorems.

Evidence freeze: 29 July 2026

Abstract

This monograph consolidates the reviewed mathematical output of a long-term Yang–Mills mass-gap research programme. Its strongest positive results are: (i) a dense-core spectral criterion converting a common diagonal semigroup rate into a gap; (ii) a finite stationary Perron–Doob/OS construction whose OS space is $L^2(\pi)$ and whose descended time translation is a positive self-adjoint Markov contraction; (iii) an exact source-visible edge-resolvent threshold; (iv) a positive-block shorting theorem showing that an upper WTR bound is structural at an already known global edge; and (v) an exact candidate-edge/shorted-Dirichlet-coercivity equivalence. The programme also proved sharp contour-incidence bounds, finite graph completeness statements, many exact $SU(2)$ coefficient identities, and a family of scoped no-go theorems for finite-complexity carriers.

None of these results locates a physical four-dimensional Wilson spectral edge, proves a lower signed connected response at that edge, derives a surviving transmuted Wilson coefficient, or completes continuum OS survival. Those four items form the exact remaining bridge for the selected route.

How to read the claims

Every result carries one of the following status classes.

[A] Exact abstract theorem.

[F] Exact finite-volume or fixed-regulator result.

[C] Conditional implication whose hypotheses remain open.

[O] Obstruction or no-go theorem under stated premises.

[K] Calibration or countermodel, often in another theory.

[S] Support result or route reduction.

[OPEN] Sufficient theorem contract with unproved hypotheses.

"Proof" means a mathematical proof of the displayed scoped statement, not a proof of the Yang–Mills target. Superseded drafts are omitted from theorem statements and listed only in the provenance appendix.

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Chapter 1

Common setup and result atlas

1.1 Operators, spectral measures, and OS notation

Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{K}$, with semigroup e^{-tH}. For $\psi \in \mathcal{K}$, write

$$L_\psi(t) = \langle \psi, e^{-tH}\psi \rangle = \int_{[0,\infty)} e^{-t\lambda}\, d\mu_\psi(\lambda).$$

In an OS reconstruction, Ω denotes the vacuum and $q(F)$ the OS class of a positive-time observable. Pointed centering means $(I - P_\Omega)q(F)$; full-ground-space centering means $(I - P_0)q(F)$.

For a retained/eliminated decomposition $P + R = I$, write a self-adjoint block operator as

$$B = \begin{pmatrix} A & C \\ D & E \end{pmatrix}, \qquad D = C^*.$$

At a proposed order bound q, set $X = qP - A$ and $H_R = qR - E$. The shorted operator is $S_P(q) = X - CH_R^{-1}D$ in the strict-tail regime.

For a source s coupled by D to a tail operator E with upper edge e^+, the source-visible edge measure is

$$\nu_s(\Lambda) = \|\mathbf{1}_\Lambda(E)Ds\|^2, \qquad F_s(u) = \nu_s([e^+ - u, e^+]).$$

1.2 Executive result atlas

ID	Result	Class	Exact boundary
DCG	Common diagonal decay on a total centered core is equivalent to a spectral gap.	A	Requires a common exponent and post-quotient totality.
OS-stat	Perron–Doob stationary finite extension has $\mathcal{H}_{\mathrm{OS}} \cong L^2(\pi)$ and positive Markov time translation.	F	New stationary finite object; no continuum edge.
WTR	Edge distribution power law controls the sandwiched resolvent threshold.	A	Upper mass law alone supplies upper growth, not a lower response.
GEWTR	Positive full block supplies upper WTR control at a known global edge.	A/S	Cannot locate the edge or prove signed lower mass.
SDC-1	$B \leq qI$ is equivalent to shorted Dirichlet coercivity, with a touching-edge form version.	A/S	Does not prove Wilson satisfies it at $q_\beta < 1$.

ID	Result	Class	Exact boundary
JCT	Compatible r-contour unions carry sharp incidence charge $2^{r-1}/(2^r-1)$.	A	Abstract incidence sharpness; Wilson realization/multiplicity remain open.
FCC	Successive semigroup norm ratios converge to the cyclic spectral edge.	A/O	Finite favorable ratios cannot bootstrap a gap.
Loop core	Finite graph loop density and continuum sigma-completeness criterion.	C/F	Actual continuum measure and OS cyclicity remain open.
Trans.	Required two-loop lattice response scale $\beta^{51/121}e^{-3\pi^2\beta/11}$.	C	Required scale, not a derived Wilson coefficient.
Finite complexity	Fixed-rank/fixed-depth/analytic architectures face leakage, reach, and flat-scale obstructions.	O	Growing, reducing, nonanalytic, stochastic, and nonlocal escapes remain.

Chapter 2

The dense-core spectral theorem

Theorem 2.1 (Vector spectral criterion, VSC-1)**.** *Let $H \geq 0$ be self-adjoint and $m > 0$. For every $\psi \in \mathcal{K}$, the following are equivalent:*

1. *there are finite C_ψ, T_ψ with $L_\psi(t) \leq C_\psi e^{-mt}$ for $t \geq T_\psi$;*
2. *$\mu_\psi([0, m)) = 0$;*
3. *$L_\psi(t) \leq \|\psi\|^2 e^{-mt}$ for every $t \geq 0$.*

For $\psi \neq 0$ these are also equivalent to $\liminf_{t\to\infty} -t^{-1} \log L_\psi(t) \geq m$.

Proof. If the spectral measure is supported in $[m, \infty)$, the all-time bound is immediate from the spectral integral. Conversely, suppose the eventual bound holds and $\mu_\psi([0, m)) > 0$. Then for some $\varepsilon > 0$ the interval $[0, m - \varepsilon]$ has positive mass a. Hence $L_\psi(t) \geq a e^{-(m-\varepsilon)t}$, contradicting $C_\psi e^{-mt}$ as $t \to \infty$. The logarithmic formulation follows by taking logarithms of positive Laplace transforms. The zero vector is handled separately and creates no logarithmic expression. □

Theorem 2.2 (Dense-core gap criteria, DCG-1 and DCG-2)**.** *Let $\mathcal{K}_0 = (\ker H)^\perp$ and let $S \subset \mathcal{K}_0$ be total. There is one $m > 0$ such that every $\psi \in S$ obeys the eventual bound of Theorem 2.1, with vector-dependent finite constants, if and only if $\operatorname{spec}(H|_{\mathcal{K}_0}) \subset [m, \infty)$.*

If $H\Omega = 0$ and $S \subset \Omega^\perp$ is total in $\Omega^\perp$, the same premise implies

$$\ker H = \mathbb{C}\Omega, \qquad \operatorname{spec}(H) \subset \{0\} \cup [m, \infty).$$

Proof. Theorem 2.1 places every vector of S in the closed spectral subspace $E_H([m, \infty))\mathcal{K}$. Totality places all of $\mathcal{K}_0$ there. The converse is the sharp all-time spectral bound. In the pointed branch, totality in $\Omega^\perp$ rules out any additional zero-energy vector, because such a vector is orthogonal to the closed positive spectral subspace. □

Corollary 2.3 (Conditional OS handoff, OS-DCG)**.** *In an OS theory, suppose the centered classes $(I - P_\Omega)q(F_\lambda)$ from an admissible positive-time core are total in $\Omega^\perp$ and their exact connected diagonal coefficients equal $\langle \psi_\lambda, e^{-tH}\psi_\lambda\rangle$. A common physical exponent m with label-dependent finite constants proves vacuum uniqueness and a gap at least m. With $(I - P_0)$ centering and totality only in $(\ker H)^\perp$, one proves a gap above the full ground space but not uniqueness.*

Remark 2.4. The unproved Yang–Mills inputs are the continuum OS construction, a fixed physical-time cutoff handoff on one cofinal sequence, a common Wilson rate, nonnull descent, and post-quotient

totality. No generator-domain or unit-sphere-uniform constant is required once these inputs are present.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle6-dense-core-spectral-criterion-v2.md;review-cycle6-dense-core-spectral-criterion-v2.md`

Chapter 3

Observable completeness and hidden sectors

Theorem 3.1 (Loop sigma-completeness criterion)**.** *Let (z_n) be bounded countably many Wilson variables and let Σ_W be their completed sigma-algebra. Rational star-polynomials in the z_n are dense in $L^2(\Sigma_W)$. Consequently they are dense in the full positive-time gauge-invariant L^2 space exactly when $\Sigma_W = \Sigma_+^{\mathrm{gi}}$ modulo null sets.*

Proof. The joint map Z takes values in a compact metrizable product. On its compact support, cylinder star-polynomials separate points and contain constants. Stone–Weierstrass gives uniform density in continuous functions; continuous functions are dense in L^2 of the push-forward measure. Pullback gives the claim, and rational coefficients are dense in the complex coefficients. □

Proposition 3.2 (OS quotient continuity)**.** *For reflection-invariant measure and reflection-positive form, $\|q(F)\|_{\mathrm{OS}}^2 \le \|F\|_2^2$. Hence raw L^2 density of bounded loop representatives descends to OS density, provided the full OS algebra and measure have already been constructed.*

Proof. Cauchy–Schwarz and reflection invariance give $|\int \overline{\theta F} F, d\mu| \le \|\theta F\|_2 \|F\|_2 = \|F\|_2^2$. Thus the quotient map is continuous and carries a dense raw core to a dense quotient core. □

Counterexample 3.3 (Hidden nonnull sector). Take $X = X_d \times \{\pm 1\}$, let all Wilson loops depend only on X_d, and use identity or tensor reflection. The function $h(x, s) = s$ is nonzero, nonnull, and orthogonal to every loop polynomial. Thus finite-graph loop completeness and reflection positivity do not imply full continuum OS cyclicity.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle7-os-loop-core-density.md;review-cycle7-os-loop-core-density.md`

Chapter 4

Sharp finite and combinatorial results

4.1 Joint compatible contour charge

Theorem 4.1 (JCT-FR)**.** *Let $X_1, \dots, X_r$ be compatible repair contours with one baseline and linear repair space. Put $V = \bigcup_i X_i$ and $D_\varepsilon = \triangle_{i:\varepsilon_i=1} X_i$ for $0 \neq \varepsilon \in \mathbb{F}_2^r$. Then for some ε,*

$$|D_\varepsilon| \geq \alpha_r |V|, \qquad \alpha_r = \frac{2^{r-1}}{2^r - 1}.$$

The constant is sharp using only incidence information. If the incidence rank is d, replace r by d.

Proof. At each face in V, the nonzero incidence functional equals one for exactly 2^{r-1} of the $2^r - 1$ nonzero vectors ε. Summing $|D_\varepsilon|$ over nonzero ε therefore gives $2^{r-1}|V|$. At least one term is no smaller than the average. Sharpness is realized by the abstract incidence system $V = \mathbb{F}_2^r \setminus \{0\}$, $X_i = \{v : v_i = 1\}$, in which every nonzero ε attains the average. □

Remark 4.2. The resulting Peierls factor is $\exp[-2\beta \cos\rho\, \alpha_r |V|]$. Connected Wilson realization, source coefficients, and multiplicity entropy are not supplied by the theorem.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-joint-center-contour-bridge-v3.md;review-cycle8-joint-center-contour-bridge-v3.md`

4.2 Finite-calibration ledger

Calibration	Exact content	Boundary
Finite Wilson endpoint covariance	Reflection-square identity, strict finite covariance, normalized spectral coefficient.	No uniform rate.
Exact-center contours	Minimal sizes 4 globally, 5 at temporal boundary, 6 in bulk; fixed-contour suppression $e^{-2\beta\lvert X\rvert\cos\rho}$.	No contour-union theorem.
Two-dimensional heat kernel	$q(R) = e^{-3g^2\ell R/8}$ and gap $3g^2\ell/8$.	Two dimensions only.
Four-dimensional $\mathbb{Z}_2$ surface model	Positive source-surface/union-entropy architecture at strong coupling.	Different theory.

Calibration	Exact content	Boundary
$SU(2)$ one-link and star integrals	Exact Bessel coefficients, hinge/star norms, adjacent-spin Haar fusion rules.	Finite cells/cylinders.
Covariance quotient toy	Source norm $\sim \sqrt{3/2}/\beta$ while serial norm remains algebraic and fusion leaks.	Not a closed carrier.
Spin-leakage Schur test	Selected residual $15/(8\beta^4)$ but omitted-leakage squared ratio tends to 6.	Finite-rank conditioning fails.

Chapter 5

Spectral ratios and source-visible edges

Theorem 5.1 (FCC spectral-ratio theorem)**.** *Let $T = e^{-RH}$, $R > 0$, and $\psi \neq 0$. Define $r_n = \|T^{n+1}\psi\|/\|T^n\psi\|$. Then r_n is nondecreasing and*

$$\lim_{n\to\infty} r_n = e^{-R \inf \operatorname{supp} \mu_\psi}.$$

An eventual bound $r_n \leq q < 1$ is equivalent to exclusion of cyclic spectral support below $-R^{-1}\log q$.

Proof. The sequence $a_n = \|T^n\psi\|^2$ is a Hausdorff moment sequence. Cauchy–Schwarz gives $a_{n+1}^2 \leq a_n a_{n+2}$, so $r_n^2 = a_{n+1}/a_n$ is nondecreasing. The nth-root principle for positive moments identifies the limit with the top of the support of $e^{-2R\lambda}$, which is $e^{-2R \inf \operatorname{supp} \mu_\psi}$ after taking square roots. □

Counterexample 5.2 (No finite bootstrap)*.* Put arbitrarily small positive spectral mass at an arbitrarily soft edge. Any fixed finite list of ratios can look favorable while the monotone sequence later rises to one. Thus finite favorable ratios do not imply a gap.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-fcc-spectral-ratio-audit-v2.md;review-cycle8-fcc-spectral-ratio-audit-v2.md`

Theorem 5.3 (Source-visible WTR threshold)**.** *Let $z = e^+ + \delta$, $\delta > 0$. Then*

$$\langle s, D^*(z-E)^{-1}Ds\rangle = \int_{[0,\infty)} \frac{dF_s(x)}{\delta + x}.$$

Assume $\|D\| \leq M$ and, near zero, $F_s(u) \leq Cu^\gamma$ uniformly. Then the resolvent is $O(1)$ for $\gamma > 1$, $O(\log(1/\delta))$ for $\gamma = 1$, and $O(\delta^{\gamma-1})$ for $0 < \gamma < 1$. An edge atom of mass a contributes a/δ. Matching lower mass laws give matching sharp orders.

Proof. Push the spectral measure forward by $x = e^+ - \lambda$. Separate a fixed far part, bounded by M^2, from $x \in [0,1]$. Stieltjes integration by parts gives

$$\int_{[0,1]} \frac{dF(x)}{\delta + x} \leq C_0 + C\int_0^1 \frac{x^\gamma}{(\delta + x)^2}\, dx.$$

The substitution $x = \delta y$ yields the three regimes. An atom at $x = 0$ appears exactly as a/δ. Reversing the inequalities under a matching lower law proves sharpness. □

Remark 5.4. An upper law gives upper growth only. It does not locate the Wilson edge or prove the lower signed connected window mass required for a physical response.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/route2-batch49-programme-reflection.md;route2-batch49-programme-reflection-independent-review.md`

Chapter 6

The finite stationary Wilson–OS theorem

Theorem 6.1 (Perron–Doob stationary OS construction)**.** *Fix a registered finite spatial $SU(2)$ lattice and $\beta > 0$. Let K be a continuous positivity-preserving, positive-semidefinite, self-adjoint, injective Wilson slab operator with Perron pair (λ_0, h), $h > 0$, normalized by $\int h^2 \, d\mu = 1$. Define*

$$P(x, dy) = \frac{K(x, y)h(y)}{\lambda_0 h(x)} \, \mu(dy), \qquad \pi(dx) = h(x)^2 \mu(dx),$$

and $T = \lambda_0^{-1} h^{-1} K h$ on $L^2(\pi)$. The stationary bi-infinite Markov chain is reflection positive across a bond. Its completed future OS space is unitarily isomorphic to $L^2(\pi)$, and time translation descends to the positive self-adjoint Markov contraction T. Every nonzero bounded one-slice gauge-invariant source survives the OS quotient.

Proof. The Doob formula and $Kh = \lambda_0 h$ show that P is Markov. Self-adjointness of K gives detailed balance with π, hence a reversible stationary chain by Kolmogorov extension. For a future observable F, set $CF = \mathbb{E}[F \mid X_1]$. The Markov property and reflection give

$$\langle q(F), q(G) \rangle_{\mathrm{OS}} = \langle CF, T \, CG \rangle_{L^2(\pi)}.$$

Because $T \geq 0$, the map $JF = T^{1/2} CF$ realizes the quotient norm and has kernel exactly the OS null space. Injectivity of the Wilson slab follows from strictly positive Peter–Weyl multipliers; hence $T^{1/2}$ has dense range and J completes onto $L^2(\pi)$. Conditional expectation intertwines one time shift with T, so the descended shift is the positive self-adjoint Markov contraction. Finally, for nonzero one-slice f, injectivity and positivity give $\langle f, Tf \rangle > 0$, proving survival. □

Remark 6.2. The stationary endpoint law $h^2 \mu$ is a newly registered finite object and is not automatically the frozen open-slab law. Compactness gives finite-volume spectral discreteness, but no regulator-uniform physical edge, continuum semigroup, Wilson-core totality, or mass gap.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/route2-batch23-os-quotient-construction-v5.md;route2-batch23-os-quotient-construction-v5-review.md`

Chapter 7

Shorting theory and the exact candidate-edge test

Theorem 7.1 (GEWTR-1: positive-block shortening)**.** *Let $B = \left(\begin{smallmatrix} A & C \\ D & E \end{smallmatrix}\right)$ be bounded self-adjoint and let $\lambda \geq \sup \operatorname{spec}(B)$. Put $X = \lambda P - A$ and $H_R = \lambda R - E$. Then there is a contraction K with*

$$C = X^{1/2} K H_R^{1/2}, \qquad D = H_R^{1/2} K^* X^{1/2}.$$

For every retained s, $Ds \perp \ker H_R$ and

$$\|H_{R,\mathrm{red}}^{-1/2} Ds\|^2 \leq \langle s, Xs \rangle, \quad \langle s, D^*(H_R + \delta)^{-1} Ds \rangle \leq \langle s, Xs \rangle.$$

Proof. The block $\lambda I - B$ is positive. The standard positive-block factorization supplies the contraction K without requiring closed range. Substituting the factorization into the quadratic form gives the reduced inverse estimate; the regularized resolvent is smaller than the reduced form. □

Proposition 7.2 (GEWTR-2)**.** *For y in the eliminated sector, the form-domain criterion is $\sup_{\delta>0} \langle y, (H_R + \delta)^{-1} y \rangle < \infty$ exactly when $y \perp \ker H_R$ and $y \in \operatorname{Dom} H_{R,\mathrm{red}}^{-1/2}$.*

Proof. Apply the spectral theorem and monotone convergence to $\int (\lambda + \delta)^{-1} d\mu_y(\lambda)$. □

Remark 7.3. GEWTR begins with the global order $B \leq \lambda I$. It cannot prove that order, locate λ, or supply a lower signed connected response.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/post-route2-wtr-shortening-theorem-v2.md;post-route2-wtr-shortening-theorem-v2-independent-review.md;post-route2-wtr-shortening-theorem-v2-erratum.md`

Theorem 7.4 (CEOE/SDC-1 strict-tail equivalence)**.** *Suppose $q > \sup \operatorname{spec}(E)$. Then*

$$B \leq qI \quad \Longleftrightarrow \quad qP - A - C(qR - E)^{-1} D \geq 0.$$

At $q = \sup \operatorname{spec}(E)$, the equivalent form statement requires $X \geq 0$, $Ds \perp \ker H_R$, $Ds \in \operatorname{Dom} H_{R,\mathrm{red}}^{-1/2}$, and

$$\|H_{R,\mathrm{red}}^{-1/2} Ds\|^2 \leq \langle s, Xs \rangle \quad \textit{for every retained } s.$$

Proof. In the strict regime, triangular congruence factorization gives

$$qI - B = \begin{pmatrix} I & -CH_R^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S_P(q) & 0 \\ 0 & H_R \end{pmatrix} \begin{pmatrix} I & 0 \\ -H_R^{-1}D & I \end{pmatrix}.$$

Congruence preserves positivity and $H_R > 0$, proving the equivalence. At the touching edge, replace the inverse by the reduced closed quadratic form. The positive-block contraction criterion is equivalent to the stated kernel, domain, and form inequality. □

Counterexample 7.5 (Bounded WTR does not establish a candidate edge). Take scalar $A = q$, $E = 0$, and $D = C = \varepsilon \neq 0$. The tail resolvent correction is finite, yet the top eigenvalue of the full 2×2 block is strictly larger than q. Thus an upper WTR estimate cannot replace SDC-1.

Remark 7.6. For a proposed physical edge $q_\beta = e^{-ma_\beta} < 1$, the first open Wilson lemma is precisely SDC-1 on the complete carrier, with uniform schedule and norm control. The theorem repackages the target-strength order; it does not make it true.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/post-route2-candidate-edge-order-equivalence-v2.md;post-route2-candidate-edge-order-equivalence-v2-independent-review.md`

Chapter 8

Wilson source, scale, and no-go results from Cycles 10–18

8.1 Exact identities and their proof mechanisms

Result	Statement and proof mechanism	Consequence
Open-index source gluing	Haar integration and intertwiners give exact gluing. The unblocked absolute character sum equals $\beta e^{\beta}/(2I_1) - 1 \sim \sqrt{\pi/2}\,\beta^{3/2}$.	Early absolute norms do not contract.
Central convolution cylinder	Peter–Weyl diagonalization gives norm $(I_2/I_1)^L$. A blocked activity with $A/\beta \to \lambda$ has exact threshold $\lambda_* \simeq 1.033465466478$.	Finite cylinder calibration only.
Recoupling	The $6j$ recoupling matrix is ℓ^2-unitary by orthogonality.	Recoupling alone creates no cancellation margin.
Carrier comparison	Telescoping normalized logs gives $-\log(\|T^{\ell} y_n\|/\|y_n\|) = \Gamma_n - c_n$.	All comparison costs must be paid.
Physical clock	A mesoscopic block $A = \Theta(\beta)$ has vanishing physical diameter. Strong continuity forces surviving response $m\tau_a + o(\tau_a)$.	Order-one response at shrinking time is invalid.
Perimeter invariance	Rescaling open kernels by z_a^{n-m} preserves positivity/gluing but shifts raw energy; separable perimeter factors cancel from centered ratios and mixed logs.	Open charged energy is not physical mass.
Finite-valence star	Haar orthogonality gives norm $c_{1/2}$ for every fixed valence; compressed norms remain algebraically close to one.	Valence does not create a transmuted margin.
Centered neutral channel	Bessel asymptotics give $-\log c_j = 2j(j+1)/\beta + j(j+1)/\beta^2 + O(\beta^{-3})$.	Wrong algebraic clock.

8.2 Diagonal-norm obstruction

Proposition 8.1 (Fixed diagonal fusion-norm no-go)**.** *For fixed β-independent diagonal weights d_j^p, the centered source norm satisfies*

$$\|S_\beta\|_p^2 \sim \frac{1}{8}\Gamma\left(\frac{p+7}{2}\right)\beta^{(p-1)/2}.$$

Source smallness requires $p < 1$, while continuous evaluation and the standard pointwise fusion algebra require $p > 3$. Hence no fixed norm in this class has both properties.

Proof. Insert the large-β, $j = O(\sqrt{\beta})$ heat-kernel/Bessel asymptotics into the weighted character sum and pass to the Gaussian Riemann integral. The stated gamma factor is the resulting radial moment. Evaluation/fusion stability requires summability of the inverse dimension weights, forcing $p > 3$, whereas the displayed power is bounded only for $p \le 1$ and small only for $p < 1$. □

Remark 8.2. The theorem excludes a fixed diagonal architecture, not non-diagonal signed blocks, connected quotients, transported norms, or scale-dependent carriers.

8.3 Dimensional transmutation and finite-rank leakage

Proposition 8.3 (Required two-loop response scale)**.** *If a finite physical $SU(2)$ mass M_F tracks the generated two-loop lattice scale, then*

$$aM_F = K_F\,\beta^{51/121}\exp\left(-\frac{3\pi^2}{11}\beta\right)[1 + O(\beta^{-1})].$$

Proof. Integrate the two-loop beta function using the universal $SU(2)$ coefficients b_0, b_1, convert $g^2 = 4/\beta$, and collect the power $b_1/(2b_0^2) = 51/121$ and exponential $1/(2b_0 g^2) = 3\pi^2\beta/11$. The physical-mass identification is the conditional premise. □

Remark 8.4. The scale is beyond every algebraic order. It is a required matching scale, not an action-derived nonzero Wilson coefficient.

Proposition 8.5 (Finite Schur recompression fails)**.** *Adding the first leaked spin can reduce one selected Schur residual to $15/(8\beta^4) + \cdots$, while the squared omitted-leakage ratio tends to 6; the full fusion Gram singular values tend to $(3 \pm \sqrt{5})/2$ and the two-fusion return remains algebraic.*

Proof. Compute the exact finite Gram and Schur matrices from adjacent-spin Haar coefficients and expand the Bessel ratios. The small selected residual is cancelled by conditioning inside the retained block, but the orthogonal leakage channel has nonvanishing norm and restores the algebraic return. □

Chapter 9

Route 2: normalized obstruction library

Route 2 used 500 identifiers for result rows, falsifiers, deferred items, and gates. The correct mathematical unit is a reviewed theorem family, not an identifier. The following table is the consolidated family ledger.

Range	Family	Proved content and escape clauses	Credit
001–010	Finite-rank leakage	Natural spin cutoffs leak; fixed-rank Wilson-filter approximation degenerates; finite analytic carriers cannot manufacture a nonzero flat response. Growing/full carriers escape.	0
011–020	Local fusion	Fixed finite spin/bond carrier and fixed depth cannot remain faithful under repeated fundamental fusion while certifying a transmuted response. Reducing cores and nonanalytic input escape.	0
021–030	Moving comparison	Metric changes redistribute response among drift, comparison, and tail return; coherent response-tempered gauges preserve the normalized exponent. Nonanalytic full-carrier telescoping escapes.	0
031–040	Tensor width	Saturated crossed singlets require exponential bond dimension; fixed bond fails on growing boundaries. Exact invariant-core states show nonuniversality.	0
041–050	Locality/reach	Finite propagation requires depth $\Omega(R/a)$ for fixed physical reach; finite analytic local data do not generate a flat scale. Rare events and growing depth escape.	0
051–060	Norm pairs	Tame strong-to-weak interpolation does not create same-space contraction; derivative loss accumulates. Genuine smoothing with same-space response escapes.	0
061–070	Cancellation complexity	Transmuted sign certification can require precision linear in β in declared numerical/oracle models. Small values need not have high description complexity.	0

Range	Family	Proved content and escape clauses	Credit
071–080	Full spin-network tests	Exact finite boundary carrier, multiplicity projector, vacuum cancellation, and finite OS positivity; naive comparison has extensive cost. No cofinal theorem.	0
081–100	Consolidation attacks	Faithful fusion depth gives dimension growth; crossed singlets give bond growth; local depth gives reach bound; analytic germs cannot be positive asymptotic to $K\beta^p e^{-A\beta}$. Premise-bounded only.	0
101–140	Operator ideals/quotients	Compact surgery cannot remove a unit essential/Calkin edge on the retained carrier; exact quotient needs invariant null space; leakage, Schur return, resolvent, Riccati, and comparison costs are explicit. Reducing/nonlinear escapes.	0
141–200	Moving carriers/categories	Wilson filter rank grows as $\Theta(\sqrt{\beta})$ at fixed tolerance; physical clock is much deeper; fusion support radius grows linearly and multiplicity exponentially; no finite complete fundamental-fusion carrier.	0
201–250	Finite Wilson/OS	Finite source/reflection kernels and the stationary theorem are exact. Open/stationary handoff, continuum binding, core totality, and physical edge remain held.	0
251–300	Proof complexity/numerics	Encoding, output-size, query, oracle, and certification bounds are exact only in their declared models. Scaling experiments cannot prove a uniform theorem.	0
301–350	Stochastic mechanisms	Rare-event, contour, polymer, and martingale identities show abstract nonanalytic mechanisms and failure modes. No compatible Wilson C/R/K/N/O package is derived.	0
351–400	Constructive architectures	Feshbach, Krylov, graph-transform, and nonuniform infinite-carrier mechanisms can realize flat scales abstractly; small residuals may accumulate order-one error. Exact Wilson coefficient/sign/schedule/OS survival remain absent.	0
401–450	Normalization/audits	Twelve scoped families normalized; contradictions, priority, significance, and publication audited. These are evidence gates, not new YM results.	0
451–500	Terminal gates	Manuscript, hostile review, reproducibility, reflection, and final audit close as typed null/nonpublication.	0

9.1 A compact publication-form obstruction theorem

Theorem 9.1 (Separated finite-complexity obstruction)**.** *Under the corresponding explicit architecture hypotheses:*

1. *a faithful linear carrier through fundamental fusion depth n has dimension at least $n+1$;*
2. *a family containing m crossed singlets needs bond dimension at least 2^m;*
3. *a direct block of radius b followed by ℓ radius-r local layers has reach at most $b + r\ell$, so fixed physical reach requires $\ell = \Omega(1/a)$;*
4. *a real-analytic response germ at $1/\beta = 0$ cannot be positive asymptotic to $K\beta^p e^{-A\beta}$ with $A > 0$;*

5. *replacing a normalized response q by $\widehat{q}$ over N steps preserves a target exponent ρ only with $|\log \widehat{q} - \log q| = o(N\rho)$ or complete two-sided accounting.*

Proof. (1) Repeated fundamental fusion reaches $n+1$ distinct spin labels. (2) Schmidt rank multiplies across independent crossed singlets. (3) Supports grow by at most r per local layer. (4) A nonzero analytic germ has a first nonzero algebraic order, while a flat exponential is smaller than every power; the zero germ cannot be positive asymptotic to it. (5) Telescope the logarithmic response difference over N steps. □

Remark 9.2. The theorem is deliberately separated. It does not exclude growing carriers, reducing sectors, nonanalytic inputs, rare-event mechanisms, nonlocal methods, or exact full-carrier constructions.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/route2-batch09-theorem-consolidation-attacks-v2.md;review-route2-batch09-theorem-consolidation-attacks-v2.md`

Chapter 10

Counterexample and barrier catalogue

Proposition 10.1 (Maxwell spin-wave method barrier)**.** *In a Gaussian Maxwell spin-wave model, let ϕ be real, smooth, compactly supported, and normalized by $\int \phi = 1$, and compare separated time translates with the declared reflection-positive ultraviolet regularization. For $X = F_{12}(\phi)$ one has $c_\beta(t) = 1/(\pi^2 \beta t^4) + O_\phi(\beta^{-1} t^{-5})$. For $W = \cos(qX)$,*

$$G_\beta(t) = e^{-q^2\sigma^2}\left(\cosh(q^2 c_\beta(t)) - 1\right) \sim \frac{q^4 e^{-q^2\sigma^2}}{2\pi^4\beta^2} t^{-8}.$$

Thus local curvature convexity, gauge fixing, Gauss projection, asymptotic stiffness, and a bounded gauge-invariant source do not by themselves imply a common exponential center-free rate.

Proof. Use the Gaussian characteristic function for the cosine covariance and expand $\cosh u - 1 \sim u^2/2$. On a cofinal sequence $\beta_j = \Theta(\log L_j)$, choose $t_j = (3/m)\log \beta_j = o(L_j)$. The polynomial tail is then incompatible with a regulator-uniform common-exponent conclusion having the Cycle-8 LC-NG premises and fixed source normalization. □

Remark 10.2. This is a method-class countermodel, not compact $SU(2)$ and not an identification of the Wilson center-free sector with Maxwell theory.

Barrier	What it rules out
Hidden soft OS sector	Tested-loop decay without post-quotient totality does not imply a full gap.
Finite favorable FCC ratios	No finite temporal window bootstraps a cyclic spectral edge.
Open-line normalization freedom	Raw charged open energy is not a physical mass without closure/normalization invariance.
Fixed diagonal fusion norm	Absolute same-space diagonal norms destroy the signed cancellations needed at weak coupling.
Finite-rank Schur conditioning	A small selected residual can coexist with order-one omitted leakage.
Calkin/essential edge	Compact perturbations cannot remove an essential unit edge on the same retained carrier.
Graph residual accumulation	A small local residual may accumulate order one over the physical clock.

Barrier	What it rules out
Exact flattening	A fully constrained finite hierarchy and its existentially eliminated endpoint relation agree on extensional endpoint queries.

Chapter 11

The exact remaining Wilson–OS bridge

The following packages are necessary for the selected route. They are not claimed necessary for every conceivable proof of the mass gap.

Open contract 11.1 (B1: physical-edge localization)**.** Construct the actual full-carrier Wilson transfer/block operator and prove SDC-1 at $q_\beta = e^{-ma_\beta} < 1$ uniformly in the declared volume/cutoff/boundary schedule. This must derive the order from the Wilson action and overlap geometry; numerical fitting or an assumed edge does not count.

Open contract 11.2 (B2: lower signed connected window mass)**.** After vacuum/connected subtraction, prove a positive lower bound on the signed source-visible spectral mass in a window adjacent to q_β. The source must be nonnull in the OS quotient and cancellation cannot be replaced by an absolute majorant that diverges.

Open contract 11.3 (B3: transmuted coefficient and physical clock)**.** Derive the nonzero Wilson coefficient at $\Theta(\beta^{51/121}e^{-3\pi^2\beta/11})$, and control leakage, comparison, and accumulated clock error at that same scale. An algebraic error overwhelms the signal and is therefore inadmissible.

Open contract 11.4 (B4: OS survival and continuum handoff)**.** Prove tightness/continuum OS reconstruction, nonnull source descent, compatible physical-time semigroup convergence, and cyclic or sigma-complete Wilson core totality. Only then may Theorem 2.2 be invoked.

11.1 Dependency chain

$$\boxed{\text{Wilson SDC-1}} \Longrightarrow \boxed{\text{physical edge}} \Longrightarrow \boxed{\text{lower signed window mass}}$$
$$\Longrightarrow \boxed{\text{transmuted response + clock}} \Longrightarrow \boxed{\text{nonnull continuum OS core}}$$
$$\Longrightarrow \boxed{\text{dense-core gap criterion}}.$$

GEWTR supplies only an upper positive resolvent estimate after the first box is already established. It does not bypass any later box.

Chapter 12

Scientific accounting and freeze verdict

Cycle-500 disposition.
PASS-RECONCILIATION / TYPED-NULL PROGRAMME CLOSURE /
ZERO YANG–MILLS CREDIT / ZERO PUBLICATION AUTHORITY.
The report contains useful mathematics and a sharper research priority. It does not contain a major Yang–Mills breakthrough or a currently authorized publishable theorem. The strongest conceptual conclusion is that ambient positive WTR upper control is structural at an already known edge; the hard physics is edge localization and lower signed connected response at the transmuted scale, followed by clock and OS survival.

The terminal evidence snapshot contains 37 unique bound rows with zero missing paths, hash/size mismatches, duplicate paths, or outside-repository entries. Its SHA-256 is `31507D5BA847AE756765EF817B3E3DFF14DA0BA1AE00F9928A0BF1BAADB3F056`. The independent frozen-snapshot review has SHA-256 `629CC9DB5258C14935577556B2D2D35B9D2C3EFD4D2C9A9ED586420A7CCC8523`.

Appendix A

Canonical provenance map

Topic	Canonical source and binding review
Dense core	`cycle6-dense-core-spectral-criterion-v2.md`; `review-cycle6-dense-core-spectral-criterion-v2.md`
Wilson core contract	`cycle6-wilson-observable-core-contract-v2.md`; matching v2 review
Loop completeness	`cycle7-os-loop-core-density.md`; matching review
Joint contour charge	`cycle8-joint-center-contour-bridge-v3.md`; matching v3 review
FCC ratios	`cycle8-fcc-spectral-ratio-audit-v2.md`; matching v2 review
Dimensional transmutation	`cycle17-dimensional-transmutation-response-scale-v2.md`; matching v2 review
Finite stationary OS	`route2-batch23-os-quotient-construction-v5.md`; `route2-batch23-os-quotient-construction-v5-review.md`
WTR threshold	`route2-batch49-programme-reflection.md`; independent review
GEWTR	`post-route2-wtr-shortening-theorem-v2.md`; independent review and rendering erratum
SDC-1	`post-route2-candidate-edge-order-equivalence-v2.md`; independent v2 review
Terminal synthesis	`route2-cycle500-final-synthesis-v2.md`; `route2-cycle500-freeze-record-v1.md`
Frozen manifest	`route2-cycle500-terminal-manifest-v5.tsv`; independent review v4

Appendix B

Supersession and correction rules

Only canonical versions named in Appendix A are used for theorem statements. In particular: Cycle 6 uses the v2 zero-vector and pointed-ground-state repair; Cycle 8 uses the v3 joint-contour baseline/onset repair and v2 FCC cutoff-edge repair; Cycle 17 uses the v2 two-loop exponent/lattice-scale repair; the finite stationary OS theorem uses Batch 23 v5; GEWTR uses v2 plus its rendering-only erratum; and CEOE/SDC-1 uses the v2 strict/touching domain split. Earlier files remain historical evidence and are not silently merged into canonical claims.

Appendix C

Open 11-lemma full-carrier contract

The full-carrier signed-block programme remains open. Its sufficient contract requires, in order: a complete carrier; a closed signed/connected source; Feshbach domain control; leakage control; a derived transmuted coefficient; comparison stability; physical-clock accumulation; reflection positivity; OS-null control; cofinal regulator uniformity; and dense-core totality. The finite-rank Schur calculation proves that satisfying a selected residual is not a substitute for these lemmas.

Appendix D

Complete Cycle 1–9 theorem ledger and proofs

This appendix restores the reviewed early result chain in theorem order. Each proof is complete at the level of the stated finite or conditional result; no entry receives continuum Yang–Mills credit.

Proposition D.1 (Endpoint-response duality (ERD-1))**.** *For the fixed regulated boundary-density and observable classes, the BR-1 response bound with constants $C, m_\partial > 0$ is equivalent, for every registered j, B, n, to*

$$\|\mathcal{R}_{j,B,n}\|_{\mathcal{O}^0_{j,n}\to\mathcal{G}^0_{j,0}} \le Ce^{-m_\partial n\ell_0}, \qquad \mathcal{R}_{j,B,n} = Q^0_{j,0}\mathbb{E}_{\mu_{j,B}}[\,\cdot \mid q_{j,0}].$$

Proof. For admissible normalized endpoint densities write their difference as a centered gauge-invariant g. Conditional expectation gives the exact pairing $\omega_h(F) - \omega_{h'}(F) = \langle g, \mathcal{R}(F - \mu F)\rangle$. The operator bound implies BR-1 by Cauchy–Schwarz. Conversely use $h = 1 + \varepsilon g$, $h' = 1$ for bounded g, cancel ε, take the Hilbert dual supremum, and extend by density. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-positive-fixed-width.md;review-cycle3-positive-fixed-width.md`

Proposition D.2 (Reachable contraction (RAC-1))**.** *Assume the still-open exact header factorization $J^*_{j,0}F = T_{j,0}\cdots T_{j,n-1}J^*_{j,n}F$. If every $T_{j,k}$ contracts the corresponding centered observable-reachable subspace by one uniform $q_{\mathrm{reach}} < 1$, then $\|\mathcal{R}_{j,B,n}\| \le q^n_{\mathrm{reach}}$ and ERD-1 holds with $m_\partial = -\ell_0^{-1}\log q_{\mathrm{reach}}$.*

Proof. Apply the restricted contraction successively from $k = n-1$ to 0; endpoint conditional expectation and centering are contractions. ERD-1 converts the resulting operator estimate into the response bound. No contraction is needed on the unreachable orthogonal complement. The exact factorization premise is not proved for the Wilson target here. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-positive-fixed-width.md;review-cycle3-positive-fixed-width.md`

Proposition D.3 (WEC-1–3 finite endpoint covariance)**.** *For the selected reflected plaquette endpoint pair in the finite Wilson measure, reflection positivity gives a transfer representation of the endpoint*

covariance. Source nonconstancy together with injectivity of the positive finite transfer gives strict positivity, and normalization yields the exact finite spectral coefficient.

Proof. Reflect the endpoint insertion into the opposite half and use the Wilson Boltzmann density to obtain the positive transfer Gram form. Peter–Weyl expansion gives strictly positive multipliers, hence an injective transfer; the selected centered source is nonzero, so its Gram minor is strictly positive. Divide by the endpoint variances for the normalized coefficient. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-wilson-endpoint-covariance.md;review-cycle3-wilson-endpoint-covariance.md`

Proposition D.4 (Active-variance hierarchy (AVE/ALP/BAV))**.** *For an active channel $A_k = P_{U_k} T_k|_{W_{k+1}}$, the exact one-layer loss is*

$$\|w\|^2 - \|A_k w\|^2 = \mathbb{E}[\mathrm{Var}(w(q_{k+1}) \mid q_k)] + \|(I - P_{U_k})T_k w\|^2.$$

These losses telescope along a propagated vector. If the accumulated scalar log losses $-\log\|A_k\|$, or the norms of a preregistered bounded-horizon block partition, exceed $m_\partial n \ell_0 - \log C_*$, then the corresponding endpoint product is at most $C_* e^{-m_\partial n \ell_0}$.

Proof. Use the conditional-expectation intertwiner and apply orthogonal Pythagoras first to the conditional-expectation defect and then to P_{U_k}. Product submultiplicativity proves ALP-1 and its bounded-horizon BAV-1 extension. One-dimensional scalar chains show optimality from layer norms alone; rotating weak directions show why bounded blocks can succeed when one-layer minima do not. The Wilson lower-loss hypotheses remain open. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-active-variance-v2.md;review-cycle3-active-variance-v2.md`

Proposition D.5 (Normalized transfer reduction (NTR-1–4))**.** *Writing the positive finite transfer as $T = \lambda_0 P$, the normalized WEC coefficient satisfies*

$$c_j = \frac{\langle u, P^D u\rangle}{d} = \frac{\langle f, M_s P^D M_s f\rangle}{\|f\|^2}.$$

If $q = q_0\phi_0 + q_\perp$, $u = u_0\phi_0 + u_\perp$, with $q_0 = \langle\phi_0, q\rangle > 0$, $d = \|f\|^2 > 0$, and centering gives $\langle q, u\rangle = 0$, define $P_\perp = (I - |\phi_0\rangle\langle\phi_0|)P(I - |\phi_0\rangle\langle\phi_0|)$, $\kappa_j = \|u_\perp\|^2/d$, $\delta_j = \|q_\perp\|/q_0$, and $\tau_j(D) = \langle u_\perp, P_\perp^D u_\perp\rangle/\|u_\perp\|^2$. Here $u_\perp \neq 0$ follows from centering and $d > 0$. Then

$$c_j = \frac{|\langle q_\perp, u_\perp\rangle|^2}{q_0^2 d} + \frac{\langle u_\perp, P^D u_\perp\rangle}{d} \le \kappa_j(\delta_j^2 + \tau_j(D)).$$

Proof. Factor the same power λ_0^N from numerator and partition function to obtain NTR-1; conjugate the boundary weights to obtain the Rayleigh form NTR-2. Resolve the identity into the Perron vector and its orthogonal complement and use the centering relation for NTR-3. Cauchy–Schwarz gives NTR-4. A two-state example shows centering need not remove the Perron term; none of $\kappa_j, \delta_j, \tau_j$ is uniformly controlled here. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-wec-rate-reduction-v2.md;review-cycle3-wec-rate-reduction-v2.md`

Proposition D.6 (WRX-1/CW-1 exact Wilson factorization)**.** *On a finite lattice split into outer region, separating blanket, and inner region, the Wilson density factors into outer, blanket, and inner action terms. Conditional integration therefore gives an exact same-measure channel identity through the blanket.*

Proof. Every plaquette belongs to the outer, inner, or blanket interaction set. Split the action accordingly, apply Fubini to the finite Haar product measure, and normalize the two conditional kernels. Their composition reproduces the original expectation exactly. No contraction estimate is used. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-wilson-raw-crosswalk.md;review-cycle3-wilson-raw-crosswalk.md`

Proposition D.7 (MDB-1/MDR-1 geometric rate conversion)**.** *For the registered cube decomposition, maximal cube depth and support distance differ by less than $2\ell_0$. Consequently an exponential depth bound converts to a support-distance bound with only a fixed multiplicative constant.*

Proof. The cube containing a support point has diameter below $2\ell_0$. Apply the triangle inequality between the support point, cube representative, and the reference boundary, then exponentiate the additive error. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle3-endpoint-bridge-v5.md;review-cycle3-endpoint-bridge-v5.md`

Proposition D.8 (PCL/PML active-localization identity)**.** *A patch conditional residual is bounded by the exact variance over its two blankets, and a Doob reveal decomposes that variance losslessly into orthogonal increments. The increments need not be local.*

Proof. Conditional expectation is the L^2 orthogonal projection onto the blanket sigma-algebra. Pythagoras gives the residual/variance identity. Revealing variables in order and applying martingale orthogonality gives the sum of increment squares. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle4-wal-active-localization.md;review-cycle4-wal-active-localization.md`

Proposition D.9 (FAR transverse extraction)**.** *The finite transverse Fourier decomposition of a centered witness has nonnegative momentum contributions. At least one real momentum component retains the total response up to the polynomial factor L^4; hence a genuine zero exponential rate survives extraction.*

Proof. Parseval decomposes the squared norm into nonnegative momentum sectors. The largest sector is at least the sum divided by the number of modes. Pair conjugate modes to obtain a real direction; polynomial factors disappear after division by physical time in an exponential rate. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle4-esw-active-slow-witness.md;review-cycle4-esw-active-slow-witness.md`

Proposition D.10 (Collar-energy spectral diagnostic)**.** *The boundary and insertion collar energies admit exact Laplace-transform representations with threshold inequalities obtained by splitting their spectral measures. Literal finite-lattice comparison costs can grow as* $\exp[O(\beta L^6)]$*, so the diagnostic alone has no cofinal authority.*

Proof. Apply the spectral theorem to the positive collar operator and split the integral below and above a threshold. Bound each piece by the extremal exponential on its interval. Direct action comparison over all affected plaquettes yields the displayed volume cost. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle4-wec-spectral-diagnostic-v2.md;review-cycle4-wec-spectral-diagnostic-v2.md`

Theorem D.11 (Fixed exact-center contour suppression)**.** *Actual lattice topology gives exact-center contour minima* 4 *globally,* 5 *at a temporal boundary, and* 6 *in the bulk. For a fixed repair-admissible contour* X*,*

$$\mu(E_X^-(\rho)) \le \exp[-2\beta|X|\cos\rho].$$

A rooted connected-family estimate follows after summing the declared repair multiplicities.

Proof. The topological minima follow by enumerating the incidence parity constraints on the lattice cells. Twist the center on X and repair along the admitted chain. Each bad plaquette gains at least $2\beta\cos\rho$ in action. Haar invariance compares the event with its repaired image and gives the bound; the rooted estimate is the union sum over the registered family. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle5-center-contour-threshold-v2.md;review-cycle5-center-contour-threshold-v2.md`

Proposition D.12 (Rare-event amplification obstructions (RAO/REA))**.** *For an event* B *of mass* q*, the centered unit vector* $h_B = (\mathbf{1}_B - q)/\sqrt{q(1-q)}$ *satisfies* $\|\mathbf{1}_B h_B\|^2 = 1 - q$*. Thus small event probability alone gives no small multiplication norm on a full centered space. Moreover, a reversible two-state chain can have stationary rare-state mass* ε *while its normalized rare-state mode has correlation* ρ^n*; taking* $\varepsilon \to 0$ *and* $-\log\rho \to 0$ *destroys any common rate.*

Proof. The first assertion is direct substitution. For the second choose transition probabilities $P(1|0) = \varepsilon(1-\rho)$ and $P(1|1) = \varepsilon + (1-\varepsilon)\rho$. Detailed balance is immediate and the normalized centered indicator is an eigenvector with eigenvalue ρ. These are abstract method countermodels, not Wilson-dynamics counterexamples. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle5-positive-curvature-coercivity.md;review-cycle5-positive-curvature-coercivity.md;cycle5-rare-event-amplification-obstruction.md;review-cycle5-rare-event-amplification-obstruction.md`

Proposition D.13 (Multiscale loss identities (MSR/MAR/MLR))**.** *For* $A_{r:t} = A_{r:s}A_{s:t}$ *and* $L_{r:s} = I - A_{r:s}^* A_{r:s}$*,*

$$L_{r:t} = L_{s:t} + A_{s:t}^* L_{r:s} A_{s:t}.$$

Restricting the earlier block to the later surviving range gives $\alpha_{r:t} \ge \alpha_{s:t} + (1-\alpha_{s:t})\alpha_{r:s|s:t}$*. For nonzero blocks the log losses satisfy* $\Lambda_{r:t} = \Lambda_{r:s} + \Lambda_{s:t} + \chi_{r,s,t}$ *with merge bonus* $\chi_{r,s,t} \ge 0$*.*

Proof. Expand $I-(A_{r:s}A_{s:t})^*(A_{r:s}A_{s:t})$ for MSR-1. Bound the product norm using the restriction of $A_{r:s}$ to $\overline{A_{s:t}W_t}$, square, and rearrange for MAR-1. Taking negative logarithms and isolating the submultiplicativity defect gives MLR-1. No scalar child data alone controls the merge bonus, and no Wilson lower bound is proved. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle5-multiscale-bypass-v2.md;review-cycle5-multiscale-bypass-v2.md`

Proposition D.14 (Observable-weighted polymer identity (OWI/OAC))**.** *For a finite weighted contour family define $N_a=\sum_X a_X\mathbf{1}_{E_X}$. Then*

$$\sum_X a_X\|\mathbf{1}_{E_X}h\|_2^2=\int N_a|h|^2d\mu,\qquad C(a)=\|M_{\sqrt{N_a}}|_{\mathcal{H}_{\mathrm{act}}}\|_{2\to2}^2.$$

Hence the desired active observable-weighted estimate is exactly a uniform Carleson/multiplication-operator bound; the scalar activity $\mathbb{E}N_a$ does not imply it.

Proof. Interchange the finite nonnegative sum with integration to obtain OWI-1. Taking the supremum of its quadratic form over nonzero active innovations is the multiplication-operator variational norm, proving OAC-1. The RAO indicator calculation shows why a first moment cannot replace the active operator bound. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle5-observable-weighted-polymer.md;review-cycle5-observable-weighted-polymer.md`

Proposition D.15 (Fixed-observable first-moment bridge)**.** *If a chosen centered observable has an exact positive source majorant, scalar first-moment leakage of that majorant controls its fixed-observable connected response. The result supplies neither a Wilson majorant nor a common rate.*

Proof. Dominate the signed insertion by the positive source measure, integrate the spectral tail once, and apply Tonelli. The estimate is labelwise and therefore does not provide the uniform common exponent required by DCG. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle7-fixed-observable-cluster-bridge-v2.md;review-cycle7-fixed-observable-cluster-bridge-v2.md`

Proposition D.16 (Finite graph all-loop density)**.** *On a finite connected graph, the star-algebra generated by characters of all closed paths, including retracing and self-intersections, is uniformly dense in continuous gauge-invariant functions.*

Proof. Fix a spanning-tree gauge to identify the orbit space with $SU(2)^r/\mathrm{Ad}$. Traces of all words separate compact orbits. The self-adjoint unital loop algebra is therefore dense by Stone–Weierstrass. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle7-os-loop-core-density.md;review-cycle7-os-loop-core-density.md`

Proposition D.17 (Exact two-dimensional heat-kernel FCC calibration)**.** *For $SU(2)$ heat-kernel lattice Yang–Mills on a spatial circle of circumference ℓ, a physical time block R acts on class functions by heat convolution,*

$$T(R)\chi_s = e^{-RE_s}\chi_s, \qquad E_s = \frac{g^2\ell}{2}s(s+1).$$

On the centered Wilson-character polynomial core the common contraction is $q(R) = e^{-3g^2\ell R/8}$, and the reconstructed finite-circle Hamiltonian has gap $3g^2\ell/8$.

Proof. Character orthogonality gives the exact heat-kernel convolution law, so all interior edges of the cylinder glue to parameter $g^2\ell R/2$. Peter–Weyl diagonalizes the transfer on the orthonormal characters. Removing χ_0, the largest eigenvalue is the $s = 1/2$ value. Character polynomials contain the complete character basis, hence form a countable dense core; iterating the diagonal estimate proves FCC with the stated common exponent. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-2d-fcc-calibration.md;review-cycle8-2d-fcc-calibration.md`

Proposition D.18 (Conditional RG kernel and source-polymer certificates)**.** *On every exactly tracked fixed-source cyclic space suppose $T_{j,R} = L_{j,R} + E_{j,R}$ with uniform bounds $\|L_{j,R}f\| \le q_0\|f\|$, $\|E_{j,R}f\| \le \varepsilon\|f\|$, and $q_0 + \varepsilon < 1$. Then FCC holds with $m_{\mathrm{RG}} = -R^{-1}\log(q_0+\varepsilon)$. Sufficient inputs are either two-sided centered-kernel Schur bounds with the same margin, or an exact source-connected polymer recursion in a norm isometric to, recovering, or one-time uniformly equivalent to the physical response norm.*

Proof. Insert $f = T_{j,R}^n\phi_{j,A}$ and apply the triangle inequality to obtain the one-step ratio $q_0 + \varepsilon$. Iteration and the even-index reflection-square identity yield exponential correlation decay. The kernel version follows from the two-sided Schur test; the polymer version applies the same estimate to the exact connected recursion. These are conditional certificates: no uniform four-dimensional Wilson bound is supplied. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-rg-entry-to-fcc-v2.md;review-cycle8-rg-entry-to-fcc-v2.md`

Proposition D.19 (Source-visible soft-mode stability lower bound)**.** *Let T_j^0, T_j be contractions and let unit source-carrier vectors f_j satisfy $\delta_j = \|(I - T_j^0)f_j\|$ and $\eta_j = \|(T_j - T_j^0)f_j\|$. Then*

$$\|T_j f_j\| \ge 1 - \delta_j - \eta_j.$$

Consequently a claimed bound $\|T_j f_j\| \le q < 1$ requires $\eta_j \ge 1 - \delta_j - q$; a perturbation vanishing on soft witnesses cannot create a uniform strict contraction.

Proof. Twice apply the reverse triangle inequality: $\|T_j^0 f_j\| \ge 1 - \delta_j$ and $\|T_j f_j\| \ge \|T_j^0 f_j\| - \eta_j$. Rearrangement gives the necessary perturbation size. The scalar example $T^0 = I$, $T = qI$ shows sharpness. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-rg-soft-mode-stability-nogo.md;review-cycle8-rg-soft-mode-stability-nogo.md`

Proposition D.20 (Multiscale source-reorganization budget)**.** *For reference transitions A_k and interacting transitions B_k,*

$$B_{N:1} - A_{N:1} = \sum_{k=1}^{N} B_{N:k+1}(B_k - A_k)A_{k-1:1}.$$

If $\|A_{N:1}f\| \geq 1-\delta$ and $\|B_{N:1}f\| \leq q$, then the post-transition-weighted local changes sum to at least $1-\delta-q$.

Proof. Telescope through hybrid ordered products to obtain the exact noncommutative identity. The reverse triangle inequality lower-bounds the product difference; the triangle inequality upper-bounds it by $\sum_k \|B_{N:k+1}\| \, \|(B_k - A_k)A_{k-1:1}f\|$. Combine the bounds. Thus many $O(1/N)$ changes may accumulate, but a vanishing total remainder cannot pay a fixed physical contraction budget. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle8-rg-multiscale-reorganization-budget.md;review-cycle8-rg-multiscale-reorganization-budget.md`

Proposition D.21 (Beta-flow/physical-response separation)**.** *For an infrared coupling flow $dg/ds = b(g) > 0$ and an exactly tracked response cocycle with contraction density $\gamma = -d\log\|u(s)\|/ds$,*

$$\frac{\|u(S)\|}{\|u(0)\|} = \exp\left[-\int_0^S \gamma(s)\,ds\right], \qquad \int_0^S \gamma(s)\,ds = \int_{g(0)}^{g(S)} \frac{\gamma(g)}{b(g)}\,dg.$$

The beta function supplies the clock, not a positive response density; equal beta trajectories can support zero or positive excitation gaps.

Proof. Integrate the logarithmic derivative and change variables using $ds = dg/b(g)$. Choosing two response cocycles with the same $g(s)$ but respectively zero and positive generators proves non-implication. A physical FCC conclusion still requires a uniform positive integrated response margin after all comparison and carrier costs. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle9-beta-response-separation.md;review-cycle9-beta-response-separation-v2.md`

Proposition D.22 (Finite $4D$ $\mathbb{Z}_2$ source-surface calibration)**.** *At strong coupling, the source-connected surface expansion has positive activities and a convergent rooted union-entropy sum within its explicit parameter region.*

Proof. Expand each plaquette Boltzmann factor into its two characters. Haar/group summation imposes even incidence away from the source, leaving surfaces with the prescribed boundary. Count rooted connected surfaces by an exponential entropy bound and sum the resulting geometric series. □

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/cycle9-z2-source-surface-calibration-v2.md;review-cycle9-z2-source-surface-calibration-v2.md`

Appendix E

Complete Cycle 10–18 exact-result ledger

Canonical result	Statement and proof	Scope/falsifier
Cycle 10 open-index gluing	Haar orthogonality and intertwiner resolution give exact source gluing; Bessel summation gives $\sum_{j>0} d_j^2 c_j \sim \sqrt{\pi/2}\beta^{3/2}$.	Absolute early cone diverges. Source: `cycle10-su2-source-character-kernel.md`; matching review.
Cycle 10 delayed onset	Telescoping the WRC log shows a Gaussian-soft UV orbit with vanishing defect cannot supply a fixed early margin; later blocks must pay the full mR budget and comparison costs.	Conditional obstruction. Source: `cycle10-wrc-delayed-onset-obstruction.md`; review.
Cycle 11 cylinder threshold	Peter–Weyl diagonalization gives $(I_2/I_1)^L$ and Laplace asymptotics give the exact $A/\beta \to \lambda$ activity threshold $\lambda_* = 1.033465466478\ldots$.	Calibrated cylinder only. Source/review: `cycle11-blocked-su2-source-norm.md`.
Cycle 11 recoupling	$6j$ orthogonality makes the recoupling matrix unitary; it preserves the ℓ^2 norm and cannot itself yield contraction.	Source: `cycle11-su2-recoupling-contraction-v2.md`; v2 review.
Cycle 12 shared link	Exact Haar decomposition gives hinge norm $c_{1/2}$, $PBP = 0$, cross norms $c_{1/2}$, compressed norm $c_{1/2}c_1$, and adjacent-spin selection.	Finite cell. Source/review: `cycle12-shared-link-su2-cell.md`, `cycle12-source-sector-composition.md`.
Cycle 12 scale matching	Strong continuity at physical time $\tau_a \to 0$ forces a surviving response $m\tau_a + o(\tau_a)$; an order-one response contradicts the semigroup limit and an algebraic raw response gives divergent effective mass.	Source: `cycle12-mesoscopic-to-physical-scale-matching-v2.md`; v2 review.

Canonical result	Statement and proof	Scope/falsifier
Cycle 13 perimeter freedom	$K_{mn} \mapsto z_a^{n-m} K_{mn}$ preserves gluing/positivity but shifts open energy; endpoint/perimeter factors cancel from centered ratios and mixed logs.	Open energy is normalization-dependent. Source: `cycle13-perimeter-renormalization-vs-mass-v2.md`; review.
Cycle 13 star norms	Haar recoupling gives norm $c_{1/2}$ at all finite valence; after one-face removal the norm is $c_{1/2}c_1$ for $k = 2$ and $c_{1/2}^3$ for $k \geq 3$. Turan's inequality shows no improved margin.	Source/review: `cycle13-finite-valence-link-star.md`.
Cycle 14 neutral WRC	The stated cumulative neutral response, common physical exponent, comparison budget, strong survival, and dense-core handoff are logically sufficient and invariant under endpoint/perimeter renormalization.	Hypotheses open. Source/review: `cycle14-neutral-renormalization-invariant-wrc.md`.
Cycle 15 centered source	Differentiating the character expansion removes the trivial coefficient; the two-source mixed log cancels vacuum and one-anchor parts. Yet the absolute majorant is at least $(3/2)\sqrt{\pi/2}\,\beta^{1/2}$.	Absolute values destroy cancellation; source: `cycle15-neutral-source-polymer-attempt.md`; review.
Cycle 16 weighted norm	Gaussian/Bessel Riemann sums yield the exact d_j^p asymptotic proved in Chapter 8; evaluation needs $p > 3$, smallness $p < 1$.	Fixed diagonal norms only; source: `cycle16-weighted-centerfree-source-norm.md`; review.
Cycle 17 connected quotient	The covariance Gram quotient has source norm $\sim \sqrt{3/2}/\beta$ but serial norm $c_{1/2}$; exact fusion generates spin 1 and then unbounded labels.	Non-diagonal toy fails closure; source/review: `cycle17-nondiagonal-connected-quotient-toy.md`.
Cycle 18 two-fusion return	Exact finite Gram/Schur algebra gives residual $15/(8\beta^4)$, leakage ratio 6, limiting singular values $(3 \pm \sqrt{5})/2$, and $-\log(c^2 q) = 7/\beta + 7/(2\beta^2) + \cdots$.	Finite-rank recompression fails; source: `cycle18-spin-leakage-schur-complement-v2.md`; v2 review.

Every table proof uses exact Haar/Peter–Weyl identities followed by stated Bessel or Laplace asymptotics; the source/review pair contains the full finite algebra. The table is a consolidation, not a replacement for its hypotheses.

Appendix F

The twelve normalized Route-2 families

Batch 41 normalized the exact Route-2 mathematics into twelve scoped families. The following statements and proof kernels are the complete normalized list.

F1. Approximation-number leakage. Fixed-rank approximation error is the corresponding singular/approximation number; Wilson-filter widths do not vanish uniformly at fixed rank. Proof: minimax characterization. Falsifier: growing rank.

F2. Calkin and ideal edge. Compact or Schatten perturbations do not remove a unit essential edge on the retained carrier. Proof: invariance of the essential spectrum in the Calkin algebra. Falsifier: exact reducing quotient.

F3. Nonlinear full-ball width. Adaptive linear information cannot beat the singular tail uniformly on a full Hilbert ball, while unrestricted data-dependent rank-one surrogates can fit one vector. Proof: adversarial orthogonal direction/minimax. Falsifier: restricted orbit.

F4. Coherent comparison and metric telescoping. Exact coherent gauges telescope, so scalar comparison losses cannot be charged repeatedly. Proof: insert adjacent comparison maps and cancel inverse pairs. Falsifier: a genuinely noncoherent two-sided defect.

F5. Locality, reach, and moving-carrier clock. Fixed microscopic depth has bounded reach and cannot by itself realize a fixed physical block as the cutoff shrinks. Proof: support induction and duration accounting. Falsifier: a proved growing depth/radius schedule with controlled clock.

F6. Fusion-graph and categorical carrier growth. Fundamental $SU(2)$ fusion reaches a linearly growing label ball with growing path multiplicity. Proof: Clebsch–Gordan recursion. Falsifier: an exact finite source-complete fusion-closed reducing carrier.

F7. Tensor/encoding width. Crossed singlets multiply Schmidt rank, preventing bounded-width exact encoding on the same generated class. Proof: Schmidt-rank multiplicativity. Falsifier: an exact bounded-width encoding.

F8. Signed-return/Feshbach completeness. A retained residual controls the full response only with a valid tail-resolvent domain, leakage, and Schur return. Proof: the block resolvent identity. Falsifier: a complete return calculation; one-way or absolute leakage is insufficient.

F9. Reducing-source-core escape. An invariant reducing source core can carry exact dynamics despite large ambient complexity. Proof: block diagonalization. This is an escape from, not an instance of, the obstructions.

F10. Analytic finite-data/nonuniform-input boundary. A nonzero normally convergent

analytic germ cannot be positive asymptotic to $K\beta^p e^{-A\beta}$. Proof: the first nonzero Taylor coefficient. Falsifier: a genuinely nonanalytic or nonuniform input provider.

F11. Finite-cutoff Wilson source/local-carrier facts. Exact finite lattice character coefficients, source identities, and local carrier algebra remain valid only at their stated cutoff and volume. Proof: Haar/Peter–Weyl calculation. Falsifier for promotion: absence of a uniform cofinal theorem.

F12. Finite stationary OS quotient/transfer facts. The Perron–Doob construction gives an exact finite stationary OS object. Proof: detailed balance, Markov factorization, and the $T^{1/2}$ quotient map. Falsifier for target use: no unitary frozen-open or continuum-limit handoff.

Canonical evidence: `.prism/physics-first-strong-cp-workflow-import-20260727/route2-batch41-theorem-normalization.md;route2-batch41-theorem-normalization-independent-review.md`

The WTR threshold, GEWTR shorting result, and SDC-1/CEOE equivalence proved in Chapters 7 and 11 are *post-Route-2 additions*. They are not relabeled as Batch-41 families and do not alter the twelve-family normalization.

The underlying canonical batch chain is: `route2-batch11-operator-ideals-v3.md`, `route2-batch12-nonlinear-approximation.md`, `route2-batch13-quotient-essential-spectrum-v2.md`, `route2-batch14-quotient-algebra-boundaries-v4.md`, `route2-batch15-moving-carriers-clock.md`, `route2-batch16-fusion-graph-growth.md`, `route2-batch17-categorical-carrier-completeness-v2.md`, `route2-batch18-signed-return-categories.md`, `route2-batch19-fusion-growth-clock.md`, `route2-batch20-category-reducing-sectors-v2.md`, `route2-batch21-finite-wilson-source-block-v2.md`, and `route2-batch23-os-quotient-construction-v5.md`; each is paired with its same-stem binding review or independent review in the request workspace.

Appendix G

Expanded provenance and review index

All paths below are relative to `.prism/physics-first-strong-cp-workflow-import-20260727/` unless prefixed by `docs/`. Review filenames are exact when the canonical file has a versioned repair; "same-stem review" means the literal source stem followed by `-review.md` or `-independent-review.md`, discoverable without a name alias.

Canonical source	Binding review
`cycle5-center-contour-threshold-v2.md`	`review-cycle5-center-contour-threshold-v2.md`
`cycle6-dense-core-spectral-criterion-v2.md`	`review-cycle6-dense-core-spectral-criterion-v2.md`
`cycle6-wilson-observable-core-contract-v2.md`	`review-cycle6-wilson-observable-core-contract-v2.md`
`cycle7-os-loop-core-density.md`	`review-cycle7-os-loop-core-density.md`
`cycle7-fixed-observable-cluster-bridge-v2.md`	`review-cycle7-fixed-observable-cluster-bridge-v2.md`
`cycle8-joint-center-contour-bridge-v3.md`	`review-cycle8-joint-center-contour-bridge-v3.md`
`cycle8-fcc-spectral-ratio-audit-v2.md`	`review-cycle8-fcc-spectral-ratio-audit-v2.md`
`cycle17-dimensional-transmutation-response-scale-v2.md`	same-stem v2 review
`cycle18-spin-leakage-schur-complement-v2.md`	same-stem v2 review
`route2-batch09-theorem-consolidation-attacks-v2.md`	`review-route2-batch09-theorem-consolidation-attacks-v2.md`
`route2-batch11-operator-ideals-v3.md`	`route2-batch11-operator-ideals-v3-review.md`
`route2-batch13-quotient-essential-spectrum-v2.md`	same-stem v2 review
`route2-batch14-quotient-algebra-boundaries-v4.md`	same-stem v4 review

Canonical source	Binding review
route2-batch17-categorical-carrier-completeness-v2.md	same-stem v2 review
route2-batch20-category-reducing-sectors-v2.md	same-stem v2 review
route2-batch23-os-quotient-construction-v5.md	route2-batch23-os-quotient-construction-v5-review.md
route2-batch49-programme-reflection.md	route2-batch49-programme-reflection-independent-review.md
post-route2-wtr-shortening-theorem-v2.md	post-route2-wtr-shortening-theorem-v2-independent-review.md
post-route2-candidate-edge-order-equivalence-v2.md	post-route2-candidate-edge-order-equivalence-v2-independent-review.md
route2-cycle500-terminal-manifest-v5.tsv	route2-cycle500-final-reconciliation-independent-review-v4.md

Appendix H

Final nonclaims

This document does not establish external novelty, literature priority, continuum existence, a Wightman theory, vacuum uniqueness in four-dimensional Yang–Mills, a positive physical spectral gap, glueball masses, confinement, or publication readiness. It is a consolidated mathematical research record whose result-bearing next step is Wilson SDC-1 at a physically scaled edge.